

Discrete optimization

Knapsack

Michel Bierlaire

Optimization: principles and algorithms



Knapsack

We review some classical combinatorial optimization problems and show how they can be modeled as a (mixed) integer linear optimization problem.

We start by the knapsack problem.

Knapsack



The knapsack problem

- ▶ Patricia prepares a hike in the mountain.
- ▶ She has a knapsack with capacity W kg.
- ▶ She considers carrying a list of n items.
- ▶ Each item has a utility u_i and a weight w_i .
- ▶ What items should she take to maximize the total utility, while fitting in the knapsack?



Modeling

Decision variables

$$x_i = \begin{cases} 1 & \text{if item } i \text{ goes into the knapsack,} \\ 0 & \text{otherwise} \end{cases}$$

Objective function

$$\max f(x) = \sum_{i=1}^n u_i x_i$$

Constraints

$$\sum_{i=1}^n w_i x_i \leq W, \quad x_i \in \{0, 1\}, i = 1, \dots, n$$

Summary

The knapsack problem is a classical combinatorial optimization problem, relevant in many contexts.

Think about an investor who needs to decide what assets to include in her portfolio.

We have shown that it can be written as an integer optimization problem.